Soham Santosh Kalgutkar

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Education

Fr. Conceicao Rodrigues College of Engineering — **2023 – Expected 2027**
B.E. in Computer Engineering, CGPA: 9.15 (up to Semester 6) — *Mumbai*

Dadar Pioneer Junior College — **2021 – 2023**
Higher Secondary School Certificate, 82%, MHT-CET: 96.767 Percentile — *Mumbai*

IES Manik Vidyamandir — **2021**
Secondary School Education (ICSE), 96% — *Mumbai*

Skills and Technologies

Languages: C++, Python, JavaScript, SQL, Java

AI/ML: PyTorch, Hugging Face, NLP, Computer Vision, XAI, TinyML

Tools & Frameworks: Git/GitHub, React, Node.js, Streamlit, MongoDB

Research Publications

TinySleepNetX | *TinyML, XAI, EEG Sleep Staging* | ICASET 2026 – Presented, Publication In Progress

- A lightweight deep learning model for single-channel EEG sleep staging, replacing recurrent networks with temporal attention to enable model interpretability and visualization.
- Optimized for Edge Computing/TinyML by engineering a frequency-sensitive feature extractor, achieving competitive accuracy on the Sleep-EDF dataset with minimal inference cost.

Indigenous Bovine Breed Classification | *Transfer Learning, ArcFace* | ICRETM 2026 – Presented, Publication In Progress

- Engineered a fine-grained breed classification system for 16 native bovine breeds using EfficientNet-B0 and ArcFace (Additive Angular Margin Loss), increasing the F1-score for visually similar breeds (Mehsana buffalo) by 39 %.
- Optimized for Edge AI deployment with a 54MB model footprint and 50ms latency, enabling offline mobile identification for the Bharat Pashudhan mission to facilitate fair livestock pricing and insurance verification.

Projects

ModelViz: PyTorch Architecture Visualizer | *TypeScript, Python, MCP, React* — **VS Code Marketplace**

- Developed ModelViz, a VS Code extension for seamless ML architecture visualization, utilizing torch.fx symbolic tracing and a custom AST-based Topological Shape Propagator for deterministic graph reconstruction.
- Architected a local-first environment via the Model Context Protocol (MCP) to ensure data privacy, featuring a Glassmorphism webview for dynamic tensor retracing and real-time input dimension modification.

MindTrack: Mental Health Sentiment Analyzer | *Python, DistilBERT, Hugging Face* — **Hugging Face**

- Fine-tuned and open-sourced a DistilBERT model for mental health sentiment analysis, achieving a 97.13 % accuracy and F1-score across a balanced dataset of 50,000 samples.
- Engineered a robust risk-detection pipeline to classify text into Normal and Risk Detected states, optimized for high-precision screening in social media and clinical text environments.

Awards & Patents

1st Place: Data Analytics Hackathon, IES MCRC College

Patent Published: Project K.A (Kitchen Automate): An IoT & AI-based Smart Kitchen Trolley